

And it is this, rather than the speed of its passing, that matters most to us about time. This is the fundamental thing about time. The secret of time lies in this slippage that we feel on our pulse, viscerally, in the enigma of memory, in anxiety about the future. This is what it means to think about time. What exactly is this *flowing*? Where is it nestled in the grammar of the world? What distinguishes the past, its having been, from the future, its not having been yet, in the folds of the mechanism of the world? Why, to us, is the past so different from the future?

Nineteenth- and twentieth-century physics engaged with these questions and ran into something unexpected and disconcerting—much more so than the relatively marginal fact that time passes at different speeds in different places. The difference between past and future, between cause and effect, between memory and hope, between regret and intention . . . in the elementary laws that describe the mechanisms of the world, there is no such difference.

HEAT

It all began with a regicide. On January 16, 1793, the National Convention in Paris sentenced Louis XVI to death. Rebellion is perhaps among the deepest roots of science: the refusal to accept the present order of things.[10] Among those who took the fatal decision was a friend of Robespierre called Lazare Carnot. Carnot had a passion for the great Persian poet Saadi Shirazi. Captured and enslaved at Acre by the Crusaders, Shirazi is the author of those luminous verses that now stand at the entrance of the headquarters of the United Nations:

All of the sons of Adam are part of one single body,
They are of the same essence.
When time afflicts us with pain
In one part of that body
All the other parts feel it too.
If you fail to feel the pain of others
You do not deserve the name of man.